CHUSHU TAN (谈楚抒)

Computational Neuroscience · Psychiatric Multi-Omics · Data Engineering & AI
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RESEARCH INTERESTS

- **Computational & Cognitive Neuroscience**: Resting-state EEG parameterization, 1/f aperiodic dynamics, Individual Alpha Frequency (IAF), functional connectivity, topological brain networks, leakage-free ML decoding.
- **Neuroimaging & Biomarker Synthesis**: Task-based fMRI, Activation Likelihood Estimation (ALE) coordinate meta-analysis, amyloid-β PET kinetic modeling (DVR vs. SUVR), multimodal CSF/plasma biomarker meta-analysis.
- **Psychiatric Multi-Omics**: Major depressive disorder, postmortem brain & peripheral blood transcriptomics, paired Illumina EPIC 850K methylation, GWAS/TWAS integration, Mendelian randomization.
- **Viral Oncology & AI for Science**: Epstein–Barr virus load stratification, NPC immune microenvironment, single-cell RNA-seq; autonomous LLM scientific agent architectures (LangChain / LangGraph).

EDUCATION

- **Master of Applied Psychology (part-time)** — *China University of Political Science and Law (CUPL)*, Beijing, China
  2021 – present | Focus: Computational Psychiatry, Neuroimaging Meta-Analysis & Cognitive Modeling
- **Master of Information Technology (M.IT)** — *Monash University*, Melbourne, Australia
  2018 – 2019 | Focus: Software Development, Distributed Database Systems, Machine Learning Algorithms
- **Bachelor of Science in Biological Sciences & Psychology (Double Major)** — *SUNY Buffalo*, NY, USA
  2011 – 2017 | Coursework: Molecular Biology, Evolutionary Genetics, Biopsychology, Biostatistics

PUBLICATIONS & WORKING MANUSCRIPTS

Peer-Reviewed Journal Articles

1. **Tan C.*** (2026). Depression and amyloid-β across CSF, PET, and plasma biomarkers: a systematic review and meta-analysis. ***Neuroscience***, 614, 82–93. DOI: 10.1016/j.neuroscience.2026.07.065. `[Solo First & Corresponding Author]`
2. Alsaadawe M.†, Radman B. A.†, ..., **Tan C.**, ..., Lyu X.* (2025). From Viral Infection to Malignancy: The Dual Threat of EBV and COVID-19 in Cancer Development. ***Viruses***, 17(9), 1195. DOI: 10.3390/v17091195. `[Co-author, PMC12474468]`

Manuscripts Under Review & Working Papers

3. **Tan C.*** (2026). Childhood Adversity and Reward-Related Brain Function in Task-Based Functional MRI: A Systematic Review and Activation Likelihood Estimation Meta-Analysis. *Under Review at* ***Child Abuse & Neglect*** (CHIABUNEG-S-26-02012). `[First & Corresponding Author]`
4. **Tan C.*** (2026). Parameterized Resting-State EEG Signatures of Acute Sleep Deprivation: Aperiodic Offset Elevation, Alpha Frequency Stability, and the Limits of Calibration-Free Decoding. *Target:* ***SLEEP*** (Draft v15, submission package ready). `[First & Corresponding Author]`
5. **Tan C.†**, Alsaadawe M.†, Hu T.†, & Lyu X.* (2026). Plasma EBV DNA Grouping Uncovers a Stage-Robust Myeloid-Inflammatory Signature in Nasopharyngeal Carcinoma. *Advanced Working Manuscript (Draft 12)*. `[Co-first Author]`
6. **Tan C.*** (2026). Reassessing Oxidative Phosphorylation Associations in Depression Using Public Transcriptomic, Methylomic and Genetic Data. *Full Pipeline Release (31 Reproducible Analysis Modules across 13 Cohorts)*. `[Lead Investigator]`

SELECTED RESEARCH & ALGORITHM PROJECTS

- **Resting-State EEG Parameterization & Sleep Deprivation Benchmark**: Analyzed OpenNeuro `ds004902` (*N* = 71, 142 sessions) via MNE-Python & FOOOF/specparam; isolated 1/f aperiodic dynamics from oscillatory peaks; developed leakage-free LOSO cross-validation ML models (SVM AUC=0.881) and 19-ch CSD/OMST graph harmonization.
- **Multimodal Neuroimaging & Coordinate Meta-Analyses**: Conducted 24-study systematic review of amyloid biomarkers in depression (*Neuroscience* 2026); engineered coordinate-based Activation Likelihood Estimation (ALE)

meta-analysis evaluating task fMRI reward processing under childhood adversity.

- **MDD–OXPHOS Multi-Omics Re-evaluation Suite (31 Pipelines)**: Architected 31 automated R/Python modules across 13 cohorts ($N > 2,000$); integrated blood & postmortem brain transcriptomics, paired Illumina EPIC 850K DNA methylation, and GWAS summary statistics to reassess mitochondrial dysfunction.
- **Plasma EBV DNA Load & NPC Immunogenomics**: Integrated bulk RNA-seq, stage-adjusted sensitivity models, immune deconvolution (CIBERSORT/MCP-counter), and single-cell RNA-seq (NPC43) to identify a tumor-intrinsic myeloid-inflammatory signature in low-load NPC.
- **Autonomous AI Agents for Scientific Discovery**: Engineered multi-agent LLM systems on LangChain & LangGraph with dynamic tool-calling for NCBI, Ensembl, and GEO APIs, automating scientific literature synthesis and workflow planning.

DATA ENGINEERING & INDUSTRY EXPERIENCE

- **Data Engineer — HSBC (via CEC outsourcing), Guangzhou** (*2026.06 – present*): Develop and maintain enterprise financial ETL pipelines using Prophecy & PySpark on Google Cloud Platform (GCP); configure Airflow orchestration and Jenkins CI/CD.
- **AI / Bioinformatics Algorithm Engineer — Hisi Medical Technology, Guangzhou** (*2026.01 – 2026.04*): Architected LLM research agents for biomedical discovery; built automated genomic pipelines combining GWAS, TWAS, and epigenetic fine-mapping.
- **Senior Data Analyst (contract) — HSBC, Guangzhou** (*2024.03 – 2025.10*): Engineered automated document parsing pipelines (OCR + NLP); integrated LLM APIs for unstructured text extraction; maintained Spark/Hive/Airflow ETL jobs.
- **Bioinformatics Engineer — Cloudna Biotech, Changsha** (*2021.11 – 2023.10*): Built automated WGS/WES analysis pipelines using Nextflow, Snakemake, and CWL with Docker; developed clinical ML platforms (Lasso, XGBoost, SVM) and DbABC resistome database.
- **Bioinformatics / Software Engineer — Xiansai / Sanji Biotech, Changsha** (*2020.06 – 2021.10*): Developed clinical NGS pipelines for variant detection; implemented relational database systems in MySQL.

DATA SCIENCE, ENGINEERING & ANALYSIS SKILLS

- **Data Science & Machine Learning**: Python (Scikit-learn, PyTorch, Pandas, NumPy, SciPy), XGBoost, LightGBM, Random Forest, SVM, Lasso/Ridge/Elastic Net, CoxPH survival modeling, SHAP explainability, Leave-One-Subject-Out (LOSO) CV, dimensionality reduction (PCA, UMAP, t-SNE).
- **Data Engineering & Cloud Infrastructure**: Apache Spark / PySpark, Apache Airflow, Hive, Hadoop, Prophecy, Google Cloud Platform (GCP), AWS, Alibaba Cloud, Docker containerization, Nextflow, Snakemake, CWL, Linux Shell scripting, Git/GitHub, Jenkins CI/CD, SQL.
- **Biomedical & Neuroimaging Analysis**: MNE-Python, FOOOF / specparam, Surface Laplacian (CSD), oPLI, OMST topological graphs, GingerALE coordinate meta-analysis, PRISMA 2020 / GRADE, Bulk RNA-seq (DESeq2, limma), scRNA-seq (Scanpy, Seurat), DNA Methylation (minfi, ChAMP), GWAS & TWAS (PLINK2, GEMMA, FUMA, SMR), TwoSampleMR, CIBERSORTx.
- **AI & Autonomous Systems**: LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Vector DBs (Chroma, FAISS), dynamic Tool-Calling, Agent state machine orchestration, Prompt Engineering, OCR + NLP text mining.
- **Languages**: English (Full professional working proficiency; US & Australian degree education, first-author English academic publications); Mandarin Chinese (Native).